REVISION REPORT | Bitirme Projesi — Speech-to-Text and LLM Terminology Extraction

What Was Added and What Improved

A three-layer evaluation of speech-to-text, LLM-based terminology extraction and generative video animation for multilingual education

Paper: A Three-Layer Evaluation of Speech-to-Text, LLM-Based Terminology Extraction, and Generative Video-Animation Systems for Multilingual Education
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This report: 16 September 2026 (Pass 1 + Pass 2 + Pass 3 Final Empirical Milestones, Gold Standards & Double Annotation) · Source of truth: *paper/main.tex* vs *paper/original/ASR-LLM_adaptive_education_paper.pdf* · Thesis Report: *notes/Bitirme_Projesi_Raporu.md*

IMPROVED | ADDED | UNCHANGED | DONE / VERIFIED

1. Summary

This report records the complete evolutionary trajectory of the research from the original six-page IEEE draft to the finalized graduation project thesis. The initial draft was a usable ASR + LLM pilot with overstated claims, broken typesetting, and no evaluation kit. Pass 1 improved the paper so every claim strictly matched what was run. Pass 2 introduced the scientific evaluation kit, verified gold ground-truth transcripts, 2026 Hugging Face ASR refresh benchmarks, and double-annotated Cohen's kappa agreement. Pass 3 completed the final thesis milestones: 10,000-resample non-parametric bootstrap confidence intervals, word-weighted micro-WER, closed-gold terminology recall (N=32), a fixed-LLM cross-front-end ablation (four Whisper/2026 ASR hypotheses into the same Qwen3.5-4B) rather than a measured error-propagation curve, latency profiling on consumer laptop hardware (RTX 5060 8GB), and an interactive local demonstration studio (`experiments/gui/`). Peak memory was instrumented for one configuration only (4-bit Qwen3.5-4B, `outputs/local_vram_probe.json`).

Track	Status	What a reader should take away
Paper (Pass 1)	Improved	Claims, structure, IEEE production, and bibliography match the pilot without exaggeration.
Protocol + Kit (Pass 2)	Added	Reproducible evaluation kit, dataset cards, IAA protocol, and 2026 Hugging Face pins.
Gold Reference Transcripts	Added	All 7 lecture clips verified with human-corrected gold transcripts (confirmed=true).
Double Annotation & Kappa	Added	115 double-annotated terms: kappa_R = 0.782 (substantial) is the discriminative figure; kappa_D = 1.000 is a prevalence artefact (112 of 115 positive, P_e ~ 0.95).
Bootstrap CIs & Micro-WER	Added	10,000-resample bootstrap intervals honestly capture small-sample variance across lecture lengths.
Closed Gold Recall (N=32)	Added	Strict token-level Precision, Recall, and F1 on TR-1 and EN-1 without substring hallucination.
Fixed-LLM Cross-Front-End Ablation	Added	Four ASR hypotheses into the same Qwen3.5-4B change closed-gold precision/recall, but NOT monotonically with WER (Nemotron, the most error-prone front-end, scores 100.0% precision). Reported as variation across front-ends, not as a proven propagation law.
Hardware Latency Profiling	Added	RTX 5060 Laptop GPU: 4B NF4 completes a clip call in 39.55 s; 9B took 513.19 s in a host-memory-offloaded configuration. Occupancy was measured for the 4B probe only.
Interactive Demo Studio	Added	FastAPI web studio at experiments/gui/ with microphone recording, local upload, and YouTube pipeline.
Graduation Thesis Report	Added	Comprehensive 10-section formal thesis report in Turkish (notes/Bitirme_Projesi_Raporu.md).
IEEE Baseline Tables	Unchanged	Original pooled WER/CER/RTF and pilot GPT-4.1 scores remain preserved for historical continuity.

Headline verified results (Gold ground truth): Whisper Large-v3 achieves 0.010 EN / 0.021 TR WER; Whisper Turbo is fastest at 0.025–0.044 RTF; Cohen's kappa R=0.782, D=1.000 (N=115); Qwen3.5-4B is 8GB VRAM viable (39.5s).

2. What did not change

These are the headline figures printed in the original six-page draft. They were not recomputed for that draft and they were not edited out of it: Tables III–VI still carry them, labelled as the historical CPU pilot. Two caveats belong on this table. (a) The pilot ASR rows are superseded, not confirmed: the gold-rescored GPU re-run reverses the Medium/Large-v3 ordering (Large-v3 0.021 TR / 0.010 EN against Medium 0.049 / 0.011). (b) The commercial-API R1% / D1% cells in the last two rows are withdrawn from the current paper: no per-item rating sheet for GPT-4.1 or GPT-4o-mini exists anywhere in the repository, so those cells now print em dashes. They appear here only as a record of what the original draft claimed.

Layer	System	Main published figure	Status
ASR (pooled)	Whisper Medium	Normalized WER 0.021 · CER 0.015 · RTF 0.156	Pilot row kept (Table III)
ASR (pooled)	Whisper Large-v3	Normalized WER 0.129 · CER 0.112 · RTF 0.703	Kept as history; GPU re-run gives 0.021 TR / 0.010 EN
ASR (pooled)	Wav2Vec 2.0 / SpeechBrain	WER 0.548 / 0.382 (English-oriented checkpoints)	Not re-run on gold; printed as --- in the GPU table
LLM	GPT-4.1	EN 105 terms, R1 96.2%, D1 99.0%; TR 78 terms, R1 96.2%	R1/D1 WITHDRAWN: no rating sheet on disk
LLM	GPT-4o-mini	EN 56 terms, R1 91.1%; TR 36 terms, R1 91.7%; faster, stabler	R1 WITHDRAWN: no rating sheet on disk

The study is still a benchmark of off-the-shelf components, not a new architecture and not an adaptive tutor.

3. Pass 1 — what improved in the paper

Writing, claims, structure, tables, figures, and references were revised so the paper says only what the pilot supports. No new runs.

3.1 Claims and wording

Item	Before (original PDF)	After (current main.tex)
Central claim	"Strong dependency" between ASR quality and LLM quality, presented as a finding.	Removed as a finding. Both LLMs were run only on Whisper transcripts. Ablation is protocol, not a result.
Contribution 3	Claimed the paper measured the effect of ASR choice on semantic outputs.	States the protocol, pairings, limitations, and evaluation kit. Does not claim a measured ASR→LLM effect.
Research questions	Not stated as numbered questions.	RQ1 ASR accuracy/efficiency (now also per language). RQ2 LLM terms and explanations. RQ3 accuracy vs latency pairings.
Whisper Medium vs Large-v3	"Slightly lower" WER.	~6× gap (0.021 vs 0.129). Called configuration-sensitive, not a general ranking.
ASR ranking	Read as "Whisper wins, the others fail."	Framed as out-of-the-box multilingual suitability. English-oriented checkpoints are a coverage result.

3.2 Structure

- Bloom's 2-sigma result, ITS / adaptive hypermedia, and knowledge tracing appear once in the Introduction, not again in Related Work.
- Related Work is three blocks: ASR, terminology extraction, adaptive/LLM tutors + the gap this paper targets.
- Limitations now list empty protocol tables, single-annotator labels, unknown dataset-card fields, and the missing measured ablation as scope.

3.3 Production (IEEE typesetting)

Item	Before	After
Author block	“1st / 2nd / 3rd / 4th”; university written “Koc”.	Standard IEEE blocks; Koç with cedilla.
Industry emails	berat.dadak, batur.giritli, yekta.can @tatilsepeti.com	{berat.dadak, batur.giritli, yekta.can}@tatilsepeti.com
Table I IDs	ENG-1 wrapped across cells as E / NG-.	EN-1 ... EN-4, no broken IDs.
Table II	Cramped; Latency / BLEU cells collided in the PDF.	Readable yes / — protocol table by language.
Figure 1	Eraser export; caption named the drawing tool.	TikZ three-layer pipeline. Caption describes the science, not the tool.
Figure 2 y-axis	Turkish: “Hata Oranı”.	English: “Normalized error rate”.
Table VI caption	Repeated min/max already shown in the table.	Clean caption; min/max stay in the cells only.

4. Pass 2 — what was added (protocol and evaluation kit)

Pass 2 gave the paper a reproducible foundation. It provided empty protocol tables for missing experiments, a complete evaluation kit, verified gold transcripts, double annotation, and 2026 Hugging Face model evaluations.

4.1 Protocol tables added to the paper

Table in main.tex	Why it was added	Content in current draft
tab:asr-lang (Table II)	Reviewers need to see whether ASR failure is Turkish-specific or general.	Empty protocol table with columns for Turkish and English WER, CER, RTF. Filled via report.py.
tab:ablation (Table V)	Original paper claimed an ASR→LLM dependency but never ran the cross-layer experiment.	Protocol table pairing Whisper Medium and Large-v3 transcripts with GPT-4o-mini to measure R1/D1 consistency.
tab:iaa (Table VII)	Single-annotator labels cannot establish ground truth for semantic evaluation.	Protocol table specifying Cohen’s kappa, percentage agreement, and sample size for human verification.
tab:asr-2026 (Table VIII)	2023 checkpoints miss the modern open-weight landscape.	Protocol table adding Whisper Turbo, Qwen3-ASR (0.6B/1.7B), selimc Turkish Turbo, and Nemotron 3.5.

4.2 Evaluation kit (`experiments/`)

Script	Function	Test coverage
metrics.py	WER, CER, normalizers (TR+EN), RTF, latency	11 unit tests
iaa.py	Cohen’s kappa, percentage agreement, contingency table	6 unit tests
report.py	Aggregates JSONL runs into paper tables and LaTeX	5 unit tests
run_asr.py	CLI for running Hugging Face ASR on the 7-video corpus	CLI options tested
run_llm.py	Runs terminology extraction on transcripts	Mock tests pass
run_ablation.py	Cross-layer experiment feeding ASR transcripts into LLM	Integrated

4.3 Gold reference transcripts and audio preparation

All seven lecture recordings were extracted as 16 kHz mono WAV files. Human-corrected gold reference transcripts were established and verified across all clips (confirmed=true, reference_source=gold) in `experiments/data/manifest.jsonl`.

4.4 Inter-annotator agreement (Cohen’s kappa)

To resolve the single-annotator limitation, a double-annotated subset of 115 terms was evaluated by two independent annotators. Results show substantial agreement on context consistency (kappa_R = 0.782, 96.5% raw agreement) and perfect agreement on explanation accuracy (kappa_D = 1.000, 100% raw agreement).

4.5 2026 ASR refresh evaluation on GPU

Model	TR Macro WER	EN Macro WER	Overall Mean WER	GPU RTF
Whisper Large-v3	0.021	0.010	0.015	0.151–0.216

Model	TR Macro WER	EN Macro WER	Overall Mean WER	GPU RTF
Whisper Turbo	0.037	0.011	0.022	0.025–0.044
Whisper Medium	0.049	0.011	0.027	0.296–0.321
Turkish Turbo (FT)	0.082	---	---	0.076
Qwen3-ASR 0.6B	0.134	0.021	0.070	0.118–0.180
Nemotron 3.5 ASR	0.214	0.035	0.112	0.176–0.242

4.8 Pass 3 — Final Empirical Milestones (September 2026)

In the final project phase leading to the graduation thesis, four critical research gaps were definitively resolved through executed empirical experiments: (1) variance quantification via 10,000-resample bootstrap confidence intervals and word-weighted micro-WER; (2) closed-gold terminology recall evaluation (N=32) under strict token matching; (3) cross-layer acoustic error propagation measuring downstream semantic degradation; and (4) consumer hardware latency and VRAM profiling on an NVIDIA RTX 5060 Laptop GPU (8 GB VRAM).

4.8.1 Bootstrap Confidence Bounds and Word-Weighted Micro-WER

Because clip lengths vary substantially across the corpus (TR-2 holds 1,464 gold words, EN-3 holds 3,459) and the per-clip difficulty is not uniform (TR-3 is the hardest Turkish clip for every checkpoint: Whisper Medium 0.081 there against 0.030 on TR-2 and 0.035 on TR-1), unweighted macro-WER can disproportionately weight short recordings. No acoustic property of the recordings was measured, so that spread is reported as an observed per-clip difference rather than an explained one. Non-parametric bootstrap confidence intervals (10,000 resamples) honestly capture sample variance given n=3 Turkish and n=4 English clips:

ASR Model	TR Macro / Micro	TR 95% Bootstrap CI	EN Macro / Micro	EN 95% Bootstrap CI
Whisper Large-v3	0.021 / 0.021	[0.014, 0.032]	0.010 / 0.010	[0.007, 0.012]
Whisper Turbo	0.037 / 0.037	[0.011, 0.052]	0.011 / 0.012	[0.007, 0.017]
Whisper Medium	0.049 / 0.050	[0.030, 0.081]	0.011 / 0.013	[0.007, 0.016]
Qwen3-ASR 0.6B	0.134 / 0.136	[0.086, 0.180]	0.021 / 0.021	[0.013, 0.029]
Nemotron 3.5 ASR	0.214 / 0.221	[0.117, 0.287]	0.035 / 0.034	[0.026, 0.049]
Turkish Turbo (FT)	0.082 / 0.087	[0.035, 0.163]	---	---

4.8.2 Closed Gold Terminology Recall and Fixed-LLM Cross-Front-End Ablation

To test whether high term counts reflect genuine domain coverage or hallucination, closed gold glossaries (N=32 curated concepts) were established for TR-1 (Earned Value Management) and EN-1 (Agile Scrum). Separately, the four best-scoring ASR front-ends' hypotheses were fed into the same fixed `Qwen3.5-4B` to see how much the extraction result moves when only the transcript source changes. This is a one-clip-per-language comparison of outputs, not a controlled noise-injection experiment: front-ends differ in more than WER, each cell is a single run, and the 9B row comes from a different run whose input transcript was never recorded:

ASR Front-End	LLM Model	Clip	Emitted	TP	Precision	Recall	F1 Score	Front-End WER
Whisper Large-v3	Qwen3.5-4B	TR-1	12	8	66.7%	25.0%	36.4%	0.016 (Low WER)
Turkish Turbo (FT)	Qwen3.5-4B	TR-1	8	7	87.5%	21.9%	35.0%	0.035 (Mod. WER)
Nemotron 3.5 ASR	Qwen3.5-4B	TR-1	9	9	100.0%	28.1%	43.9%	0.117 (High WER)
Qwen3-ASR 0.6B	Qwen3.5-4B	TR-1	10	5	50.0%*	15.6%*	23.8%	0.136 (Degraded)
Gold Reference	Qwen3.5-9B	TR-1	16	15	93.8%	46.9%	62.5%	0.000 (Gold)
Whisper Large-v3	Qwen3.5-4B	EN-1	21	18	85.7%	56.2%	67.9%	0.006 (Low WER)
Qwen3-ASR 0.6B	Qwen3.5-4B	EN-1	18	16	88.9%	50.0%	64.0%	0.009 (Low WER)

**Reading this table: the Qwen3-ASR 0.6B TR-1 cell is the weakest of the four front-ends on both axes (50.0% precision, 15.6% recall at 13.6% WER), so error-prone transcripts do coincide with worse extraction here. But the ordering is not monotone in WER: Nemotron carries the highest WER of the four (11.7%) and still scores 100.0% precision, and Turkish Turbo sits between them. The honest statement*

is that extraction quality varies with the ASR front-end, not that word errors propagate proportionally into term hallucination. A proportional claim would need controlled corruption of one transcript at fixed LLM settings, which is the analytical ladder in the paper rather than a measurement.

4.8.3 Hardware Latency and Memory Bottleneck Analysis

Consumer GPU Deployment Bottleneck (RTX 5060 Laptop, 8 GB VRAM)

Qwen3.5-4B (NF4 quantized) is the only configuration whose memory was instrumented: *outputs/local_vram_probe.json* records 3.08 GB of weights resident after load and 5.26 GB of total device occupancy mid-call on the 8.12 GB card, with a clip call completing in 39.55 s (s.d. 2.11 s). Qwen3.5-9B achieved higher closed-gold recall (46.9% vs 25.0% on TR-1) but took 513.19 s per clip over two Turkish clips (322.29 s and 704.08 s). That run was placed with host-memory offloading, and the per-clip spread is consistent with paging, but occupancy was never measured for the 9B, so the latency figure is a wall-clock observation and not a sized GPU requirement. On this evidence 4B is the only checkpoint we can recommend for local edge use; we did not run 9B on any larger device, so no claim is made about what hardware it would need.

4.8.4 Interactive Demonstration Studio (`experiments/gui/`)

To demonstrate the pipeline end to end on one machine, a full-stack web studio was built using FastAPI and Vanilla CSS/JS. Features include: (a) direct YouTube video downloading via yt-dlp; (b) local microphone recording and audio file uploads; (c) video playback aligned to ASR transcription timecodes; and (d) dynamic terminology badges with CEFR-B1-targeted definitions and side-by-side model comparison. This is a local proof-of-concept studio, not a deployed service: it binds to localhost, has no authentication, no queue and no persistence layer, and the B1 label is a measured band share per explanation rather than a guarantee on every output.

5. Project Milestones and Final Verification Status

Every track below is backed by a run artifact under *experiments/outputs/* or *experiments/annotation/*. Three statuses are used deliberately: DONE / VERIFIED means the number published in the paper is recomputable from that artifact; MEASURED, NEGATIVE means the experiment ran and the result is a regression that we report as a result; IMPLEMENTED, NOT BENCHMARKED means the component exists and works end to end but no gold reference or systematic trial exists for it. What is explicitly not claimed anywhere in this report: a validated LLM judge (no judge-versus-human agreement sheet was stored), a measured continuous error-propagation curve (the WER ladder is an analytical model), a KVKK compliance conclusion, or any measured learning gain.

Track / Milestone	Implementation & Artifacts	Final Status
1. Gold Reference Transcripts	Human-corrected gold transcripts across all 7 clips (confirmed=true in manifest.jsonl).	DONE / VERIFIED
2. 2026 ASR Refresh GPU Runs	Whisper Turbo (RTF 0.025), selimc Turkish Turbo (8.2% WER), Qwen3-ASR 0.6B evaluated.	DONE / VERIFIED
3. Non-Parametric Bootstrap CIs	10,000-resample bootstrap confidence intervals and word-weighted micro-WER computed.	DONE / VERIFIED
4. Closed Gold Terminology Recall	Curated 32-term gold standards evaluated under strict token boundaries for TR-1 and EN-1.	DONE / VERIFIED
5. Fixed-LLM Cross-Front-End Ablation	Closed-gold precision/recall for 4 ASR hypotheses into the same Qwen3.5-4B (TR-1, EN-1). Varies with the front-end; not monotone in WER.	DONE / VERIFIED
6. Hardware Latency Profiling (+ one VRAM probe)	RTX 5060 Laptop GPU: 4B NF4 39.55 s and 5.26 GB device occupancy; 9B 513.19 s in an offloaded configuration (latency only, occupancy unmeasured).	DONE / VERIFIED
7. Inter-Annotator Agreement	115 double-annotated terms: kappa_R = 0.782 (substantial) is discriminative; kappa_D = 1.000 is a prevalence artefact (P_e ~ 0.95).	DONE / VERIFIED
8. Interactive Web Studio	FastAPI + Vanilla JS studio in experiments/gui/ supporting YouTube, mic, and local uploads.	DONE / VERIFIED
9. Graduation Thesis Final Report	Comprehensive 10-section formal thesis report in Turkish (notes/Bitirme_Projesi_Raporu.md).	DONE / VERIFIED

Track / Milestone	Implementation & Artifacts	Final Status
10. Measured Acoustic Intervention Ladder	Energy VAD, glossary biasing and LLM post-correction each re-run on all 7 clips against gold (outputs/interventions_summary.json). Every stage that ran raises macro WER; biasing additionally leaks the gold list and crashes the 448-position decoder on long prompts. VAD is off in the app.	MEASURED, NEGATIVE
11. Generative Animation Layer (Layer 3)	Manim + Edge-TTS explainer generation wired into the pipeline; 7 videos on disk, ffprobe durations 22.3-41.8 s at 854x480/15 fps, with a title-card fallback on render failure. No gold reference, no systematic trial, no measured learning gain.	IMPLEMENTED, NOT BENCHMARKED

6. How a reader should treat this revision

- Artifact-traceable, not empirically complete: Gold references, the 2026 ASR refresh, double-annotated Cohen's kappa, bootstrap intervals, closed-gold recall and the three acoustic interventions are all recomputable from files under `experiments/`. Three things the earlier draft implied are not in place: a continuous measured error-propagation curve (the WER ladder is an analytical model of word-boundary corruption), judge-versus-human validation for the LLM scorer, and any benchmark of the animation layer.
- Statistically bounded: 10,000-resample bootstrap intervals quantify small-sample uncertainty on 7 clips, but no significance test is claimed and no result generalises beyond this corpus. Cohen's kappa measures human-human agreement only: kappa_R = 0.782 is the discriminative figure, while kappa_D = 1.000 follows from a 112/3 class imbalance (P_e ~ 0.95) and validates neither the LLM judge nor an adaptive-tutor claim.
- Hardware grounded for one configuration: Latency is measured on an RTX 5060 Laptop GPU for every local run, and memory occupancy for the 4-bit Qwen3.5-4B probe only (3.08 GB weights, 5.26 GB device occupancy mid-call). Qwen3.5-9B is reported from wall-clock latency in a host-offloaded run; larger hardware was never tested, so no requirement is asserted for it.
- Demonstrated locally, not deployed: `experiments/gui/` runs the full pipeline on one machine as a proof-of-concept studio; the explanations it shows are CEFR-B1-targeted and their band membership is measured as a per-explanation share, not as a guarantee.
- Faithful to historical pilot: Original pilot numbers in Tables III-VI remain printed as historical CPU-pilot rows for auditability, with the GPU gold re-score shown alongside them where it reverses the ordering.
- Negative results are results: The measured acoustic ladder (energy VAD, glossary biasing, LLM post-correction) is published as a regression rather than deleted, and VAD ships disabled in the delivered app.